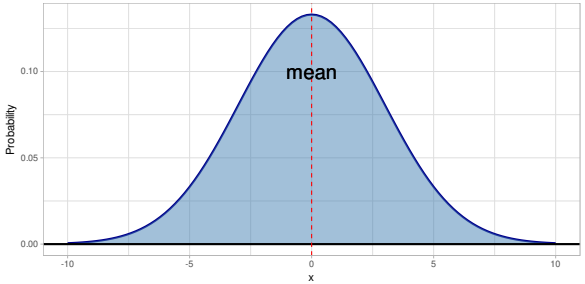
# Distribution of Measured Variables

Pietro Franceschi | pietro.franceschi@fmach.it

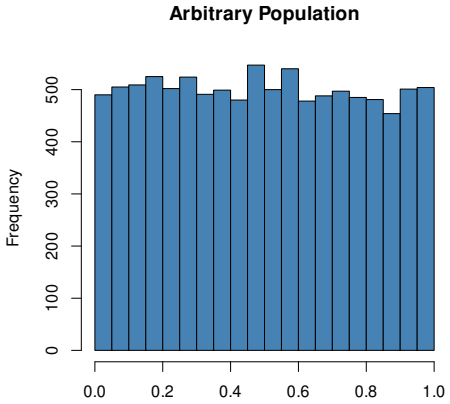
# Normal Distribution

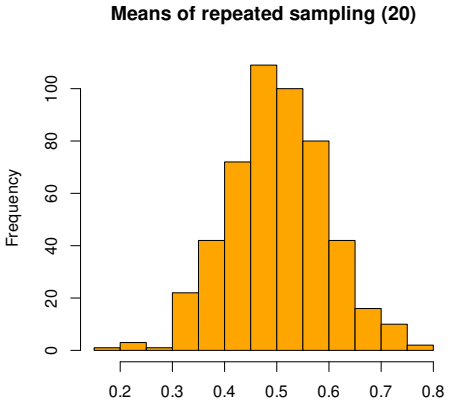
- The Normal (Gaussian) distribution has a prominent role in statistics
- Some sort of "normality" is often the prerequisite of many statistical tools
- With normal data, the mean is the value with higher probability
- Reasoning on the mean is equivalent is reasoning on the most probable value ...



# Sampling Distribution

The distribution of the mean of a sample extracted from every type of distribution is always normal



# Sampling Distribution

**Noteworthy**

1. The sampling distribution get's narrower as the sample increases

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**Noteworthy**

1. The sampling distribution get's narrower as the sample increases
2. This is the ultime reasons why measuring more samples leads to smaller *p-values*

# Formally

- Mean of the sampling distribution: $\mu_{pop}$
- Variance of the sampling distribution: $\frac{\sigma}{\sqrt{(N)}}$

Where $N$ is the size of the sample

# ... but

We are often dealing with data non normally distributed

1. Count based technologies never return negative numbers ... (MS!)

Sub populations are typically present in complex experiments

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Sub populations are typically present in complex experiments

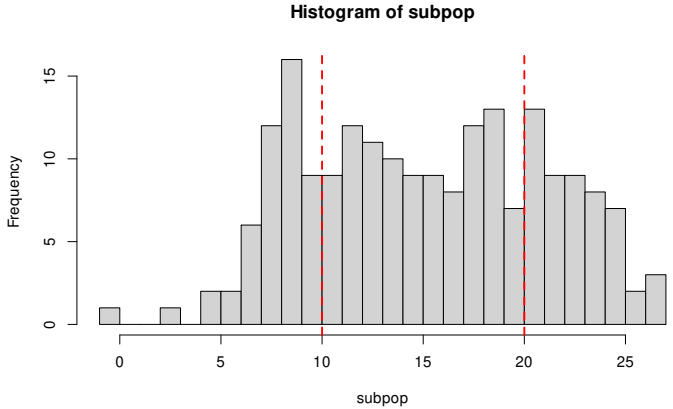
# ... but

We are often dealing with data non normally distributed

1. Count based technologies never return negative numbers ... (MS!)
2. In many biochemical reactions the product of concentrations is important
3. Presence of sub-populations
4. Outliers

Sub populations are typically present in complex experiments

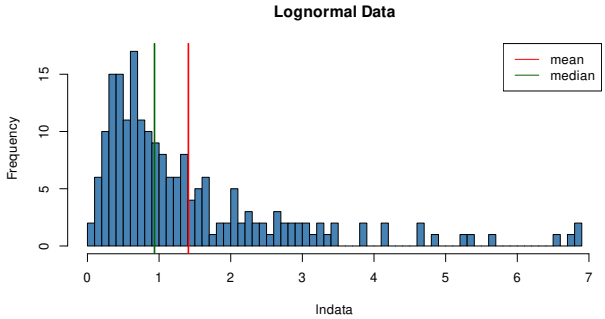
# Subpopulations




**NB:** in many cases subpopulations are unknown!

# Lognormal data

Lognormal data are extremely common in metabolomics ...




- The mean is **not** the most probable value!
- Statistical machinery focusing on **mean** is not the right tool for the trade

# Solutions

1. **Non-parametric** approaches (Kruskall-Wallis, quantile-regression, permutations, bootstrap, ...)

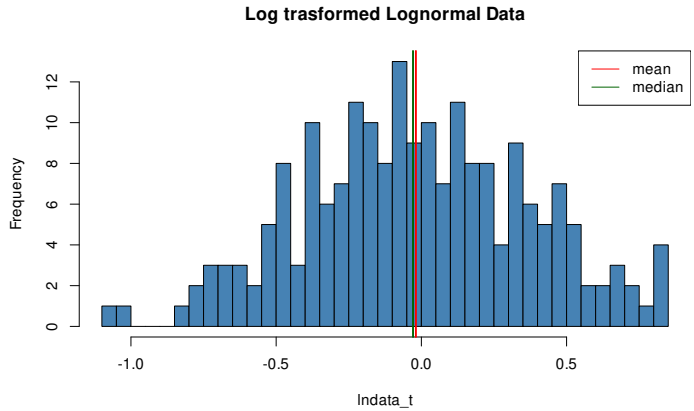
Remember that non parametric tests suffer of lack of power and are often completely useless fin investigations with only few samples (5-10)

# Solutions

1. **Non-parametric** approaches (Kruskall-Wallis, quantile-regression, permutations, bootstrap, ...)
2. Variable **transformations**

Remember that non parametric tests suffer of lack of power and are often completely useless fin investigations with only few samples (5-10)

# Log-transformed data



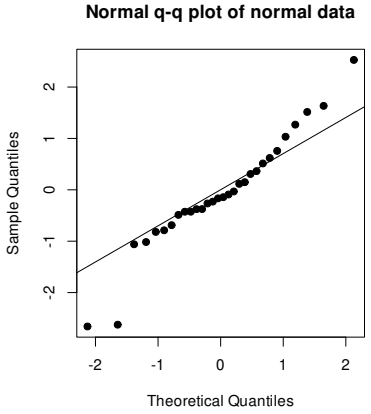

Mean and median are ok!

# Checking Normality

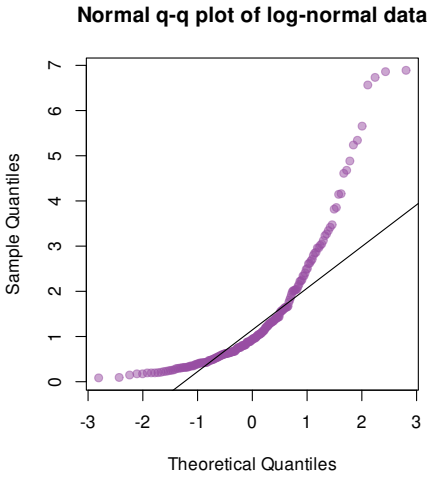
1. Normality tests ... again statistics ;-)
2. Graphical methods `q-q plots`

# Quantile - quantile plots

Quantile quantile plots are used to visually compare the **theoretical quantiles** of a distribution with the **sample quantiles**



# q-q plot for lognormal data



# Take home message

Before running your statistical machinery give a look to your data to check if what you are doing makes sense ...